

- 3 (a) The thermodynamic temperature of an ideal gas is proportional to the mean kinetic energy of the molecules of the gas.

(b) (i)
$$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

$$\frac{1}{2} \times \frac{0.030}{6.02 \times 10^{23}} \times \langle c^2 \rangle = \frac{3}{2} \times 1.38 \times 10^{-23} \times (27 + 273.15)$$

$$c = 499 \text{ m s}^{-1}$$

(ii)
$$PV = nRT$$

$$1.0 \times 10^5 \times 0.029 = n \times 8.31 \times (27 + 273.15)$$

$$n = 1.16$$

- (iii) The root-mean-square speed of the molecules increases with temperature.

Since the volume of the oven remains the same, there is an increase in the number of collisions per unit time of the molecules and the walls of the oven, causing an increase in pressure.

Or

For each collision between a molecule and the wall of the oven, there is a greater change in momentum of the molecule. This leads to a greater force exerted on the wall and an increase in pressure.

(iv)
$$PV = nRT$$

$$1.0 \times 10^5 \times 0.029 = n_2 \times 8.31 \times (220 + 273.15)$$

$$n_2 = 0.7076$$

$$\text{mass of air} = (1.163 - 0.7076) \times 0.030 = 1.37 \times 10^{-2} \text{ kg}$$